

Computer network1 Project

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1. You must set the hostname of the switches to Birthplace_CentAS1 – 4 and the hostname of the routers to Birthplace_CentR1 – 2 (for example, if you were born in Pécs, then Pecs_CentAS1 – 4 and Pecs_CentR1 – 2)!

```
Switch>enable
```

```
Switch#conf t
```

Enter configuration commands, one per line. End with CNTL/Z.

```
Switch(config)#hostname Osaka_CentASW0
```

```
Osaka_CentASW0-4(config)#
```

2. Set the enable password and password encryption on all switches and routers. (The password for the switches should be CentSwitch1 – 4 and the password for the router should be CentRouter1 – 2!)

```
Osaka_CentASW0-4(config)#enable password CentSwitch0
```

```
Osaka_CentASW0-4(config)#service pas
```

```
Osaka_CentASW0-4(config)#service password-encryption
```

3. Since secure operation is a priority, set SSH on all switches and routers. (The domain, in which the devices are located is company.hu.) To increase security, use SSH version 2! and

4. To increase security, user-based authentication is required, so create a user with an Admin and a NormalUser ID with the appropriate rights and settings, on all devices! (Passwords: Admin1 – 4: Administrator1 – 4, NormalUser1 – 4: user1 – 4) and

5. Set the accesses! Use user-based authentication for both the terminal and the console access! Only accept the SSH-based communication on the terminal connections!

```
Osaka_CentASW0-4(config)#ip domain-name company.hu
```

```
Osaka_CentASW0-4(config)#username Admin0 privilege 15 secret Administrator0
```

```
Osaka_CentASW0-4(config)#username NormalUser0 secret user0
```

```
Osaka_CentASW0-4(config)#crypt
```

```
Osaka_CentASW0-4(config)#crypto key generate rsa
```

The name for the keys will be: Osaka_CentASW0-4.company.hu

Choose the size of the key modulus in the range of 360 to 4096 for your

General Purpose Keys. Choosing a key modulus greater than 512 may take a few minutes.

How many bits in the modulus [512]: 1024

```
Osaka_CentASW0-4(config)#ip ssh version 2
```

```
*Mar 1 1:1:14.892: %SSH-5-ENABLED: SSH 1.99 has been enabled
```

```
Osaka_CentASW0-4(config)#line vty 0 15
```

```
Osaka_CentASW0-4(config-line)#login local
Osaka_CentASW0-4(config-line)#transport input ssh
Osaka_CentASW0-4(config-line)#exit
Osaka_CentASW0-4(config)#
Osaka_CentSW0(config)#line console 0
Osaka_CentSW0(config-line)#login local
Osaka_CentSW0(config-line)#exit
show startup config or show ip ssh
```

6. For safe operation, you must create the following VLAN structure on all switches!

7. Configure the VTP protocol on the switches! Configure the first two switches as a VTP server (a) and the second two as a client (a)! (In the case (b) reverse!) (The name of the VTP domain should be company, the password should be: companyvtp!)

For switch0 and 1

```
Osaka_CentSW0(config)#vtp domain company
Changing VTP domain name from NULL to company
Osaka_CentSW0(config)#vtp pass
Osaka_CentSW0(config)#vtp password companyvtp
Setting device VLAN database password to companyvtp
Osaka_CentSW0(config)#vtp mode client
Setting device to VTP CLIENT mode.
show vtp status
```

For switch2 and 3

```
Osaka_CentASW2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Osaka_CentASW2(config)#vtp domain company
Changing VTP domain name from NULL to company
Osaka_CentASW2(config)#vtp pas
Osaka_CentASW2(config)#vtp password companyvtp
Setting device VLAN database password to companyvtp
Osaka_CentASW2(config)#vtp mode server
Device mode already VTP SERVER.
Osaka_CentASW2(config)#vlan 10
Osaka_CentASW2(config-vlan)#name Operation
Osaka_CentASW2(config-vlan)#exit
Osaka_CentASW2(config)#vlan 20
Osaka_CentASW2(config-vlan)#name HR
Osaka_CentASW2(config-vlan)#exit
Osaka_CentASW2(config)#vlan30
^
```

% Invalid input detected at '^' marker.

```
Osaka_CentASW2(config)#vlan 30
Osaka_CentASW2(config-vlan)#name Manufacturing
```

```

Osaka_CentASW2(config-vlan)#exit
Osaka_CentASW2(config)#vlan 40
Osaka_CentASW2(config-vlan)#name EcoOff
Osaka_CentASW2(config-vlan)#exi
Osaka_CentASW2(config)#vlan90
^

```

% Invalid input detected at '^' marker.

```

Osaka_CentASW2(config)#vlan 90
Osaka_CentASW2(config-vlan)#name MGMNT
Osaka_CentASW2(config-vlan)#exit
Osaka_CentASW2(config)#vlan 100
Osaka_CentASW2(config-vlan)#name Spare
Osaka_CentASW2(config-vlan)#exi
Osaka_CentASW2(config)#exit
Osaka_CentASW2#wr
show vlan brief

```

8. For proper operation, divide the FastEthernet ports of the switch between VLANs as follows: (In each VLAN, end-devices are connected to the ports!) VLAN 1X: 1 – 3 port (a), 24 – 21 (b), VLAN 2X: 4 – 7 port (a), 20 – 17 (b), VLAN 3X: 8 – 11 port (a), 16 – 13 (b), VLAN 4X: 12 – 15 port (a), 12 – 9 (b). to increase the security, move unused ports to Other (a) or Spare (b) VLANs and shutdown them. enable bpduguard service on access ports! Additionally, set the port security on the ports you are using, with the following parameters: Sticky MAC address management, maximum 3 address.

* At the process 12, I need to allocate the ports for trunk. so unused ports are 2-8.

Before configuring access ports for each Vlan, the vlan of client switches should be shared from the sever switches. So process 12 should be done firstly.

12

```

Osaka_CentASW3(config)#interface fa0/1
Osaka_CentASW3(config)#switchport mode trunk
Osaka_CentASW3(config)#switchport trunk native vlan 90
Osaka_CentASW3(config)#switchport trunk allowed vlan 10,20,30,40,90,100
Osaka_CentASW3(config)#no shutdown
Osaka_CentASW3(config)#exit
show interface trunk
show vlan brief

```

8

```

Osaka_CentSW0(config)#interface range fa0/21 - 24
Osaka_CentSW0(config-if-range)#switchport mode access
Osaka_CentSW0(config-if-range)#switchport access vlan 10

```

```
Osaka_CentSW0(config-if-range)#span
Osaka_CentSW0(config-if-range)#spanning-tree portfast
Osaka_CentSW0(config-if-range)#spanning-tree bpdug
Osaka_CentSW0(config-if-range)#spanning-tree bpduguard enable
Osaka_CentSW0(config-if-range)#switch
Osaka_CentSW0(config-if-range)#switchport por
Osaka_CentSW0(config-if-range)#switchport port-security
Osaka_CentSW0(config-if-range)#switchport port-security mac-addre
Osaka_CentSW0(config-if-range)#switchport port-security mac-address sti
Osaka_CentSW0(config-if-range)#switchport port-security mac-address sticky
Osaka_CentSW0(config-if-range)#switchport port-security maxi
Osaka_CentSW0(config-if-range)#switchport port-security maximum 3
exit
Osaka_CentSW0(config)#interface range fa0/17 - 20
Osaka_CentSW0(config-if-range)#swit
Osaka_CentSW0(config-if-range)#switchport mode acc
Osaka_CentSW0(config-if-range)#switchport mode access
Osaka_CentSW0(config-if-range)#switchport access vlan 20
Osaka_CentSW0(config-if-range)#span
Osaka_CentSW0(config-if-range)#spanning-tree portfast
Osaka_CentSW0(config-if-range)#spanning-tree bpd
Osaka_CentSW0(config-if-range)#spanning-tree bpduguard enable
Osaka_CentSW0(config-if-range)#sitc
Osaka_CentSW0(config-if-range)#swi
Osaka_CentSW0(config-if-range)#switchport port
Osaka_CentSW0(config-if-range)#switchport port-security
Osaka_CentSW0(config-if-range)#switchport port-security mac-
Osaka_CentSW0(config-if-range)#switchport port-security mac-address stic
Osaka_CentSW0(config-if-range)#switchport port-security mac-address sticky
Osaka_CentSW0(config-if-range)#switchport port-security maxi
Osaka_CentSW0(config-if-range)#switchport port-security maximum 3
Osaka_CentSW0(config-if-range)#exi
Osaka_CentSW0(config)#interface range fa0/13 - 16
Osaka_CentSW0(config-if-range)#switchport mode access
Osaka_CentSW0(config-if-range)#switchport access vlan 30
Osaka_CentSW0(config-if-range)#spanning-tree portfast
Osaka_CentSW0(config-if-range)#spanning-tree bpduguard enable
Osaka_CentSW0(config-if-range)#switchport port-security
Osaka_CentSW0(config-if-range)#
Osaka_CentSW0(config-if-range)#switchport port-security mac-address sticky
Osaka_CentSW0(config-if-range)#switchport port-security maximum 3
Osaka_CentSW0(config-if-range)#exi
Osaka_CentSW0(config)#interface range fa0/9 - 12
Osaka_CentSW0(config-if-range)#switchport mode access
Osaka_CentSW0(config-if-range)#switchport access vlan 40
Osaka_CentSW0(config-if-range)#spanning-tree portfast
Osaka_CentSW0(config)#interface range fa0/4 - 8
Osaka_CentSW0(config-if-range)#switchport mode access
```

```
Osaka_CentSW0(config-if-range)#switchport access vlan 100
Osaka_CentSW0(config-if-range)#shutdown
show vlan brief
show port-security interface fa0/21
show port-security
```

9. Configure the spanning-tree protocol on the switches with the following parameters: Set the portfast service on the access ports on all switches! Set the first switch as the primary root switch for 1X, 2X, and 3X VLANs, and the secondary root switch for 4X and 9X VLANs. In (b) use the third switch instead of the first! Set the second switch as the secondary root switch of the 1X, 2X, and 3X VLANs, and to the primary root switch of the 4X and 9X VLANs. In (b) use the fourth switch instead of the second!

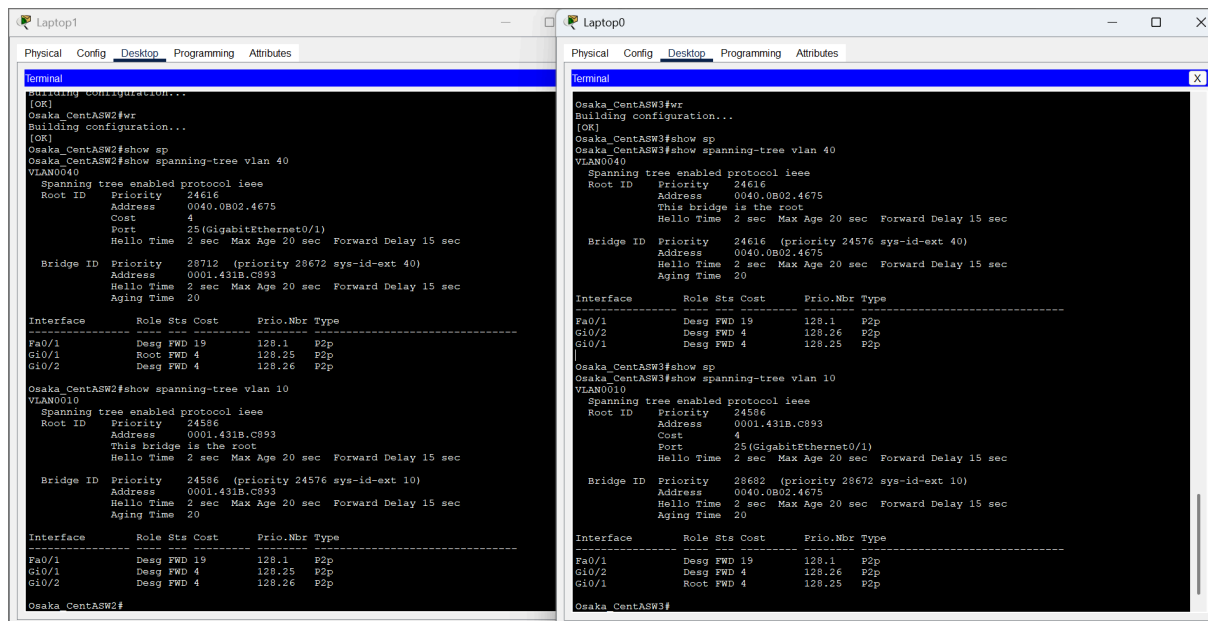
switch 3

```
Osaka_CentASW3(config)#spanning-tree vlan 10 root secondary
Osaka_CentASW3(config)#spanning-tree vlan 20 root secondary
Osaka_CentASW3(config)#spanning-tree vlan 30 root secondary
Osaka_CentASW3(config)#spannin
Osaka_CentASW3(config)#spanning-tree vlan 40 root primary
Osaka_CentASW3(config)#spanning-tree vlan 90 root primary
Osaka_CentASW3(config)#exi
```

switch 2

```
Osaka_CentASW2(config)#spann
Osaka_CentASW2(config)#spanning-tree vlan 10 root primary
Osaka_CentASW2(config)#spanning-tree vlan 20 root primary
Osaka_CentASW2(config)#spanning-tree vlan 30 root primary
Osaka_CentASW2(config)#spanning-tree vlan 40 root secondary
Osaka_CentASW2(config)#spanning-tree vlan 90 root secondary
Osaka_CentASW2(config)#exi
```

```
show spanning-tree vlan 10
show spanning-tree vlan 40
```



10. Set the SVI for the Management (a) (MGMNT (b)) VLAN on each switch, to the IP address from 192.168.9X.247 to 192.168.9X.250!

11. Set the default gateway address of the switches to 192.168.9X.254!

```
Osaka_CentSW0(config)#interface vlan 90
Osaka_CentSW0(config-if)#
%LINK-5-CHANGED: Interface Vlan90, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan90, changed state to up
Osaka_CentSW0(config-if)#ip address 192.168.90.247 255.255.255.0
Osaka_CentSW0(config-if)#no shutdown
Osaka_CentSW0(config-if)#exi
Osaka_CentSW0(config)#ip default-gateway 192.168.90.254
show ip interface brief
```

13. Configure the routing between VLANs, on routers! (Use the Router-on-a-Stick method! The ID of the sub-interfaces must match the ID of the VLAN! The IP addresses must be 192.168.VLAN_ID.252 on CentR1 and 192.168.VLAN_ID.253 on CentR2, with the appropriate subnet mask!)

```
Osaka_CentR0(config)#interface fa0/0
Osaka_CentR0(config-if)#no shut
Osaka_CentR0(config)#interface fa0/0.10
Osaka_CentR0(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/0.10, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.10, changed state to up
Osaka_CentR0(config-subif)#enca
Osaka_CentR0(config-subif)#encapsulation dot1Q 10
Osaka_CentR0(config-subif)#ip address 192.168.10.252 255.255.255.0
Osaka_CentR0(config-subif)#exi
```

and then configure for each vlan
show ip interface brief

14. Configure the HSRP with load-sharing on routers, with the following parameters: CentR1 should be the active router in 40 and 90 VLANs, and standby in 10, 20 and 30 VLANs. CentR2 should be the active router in 10, 20 and 30 VLANs, and standby in 40 and 90 VLANs! The IP address of active routers should be 192.168.VLAN_ID.254, and the IP address of standby routers should be 192.168.VLAN_ID.1! Standby routers must be able to take over the active role if necessary, and active routers must be able to take back this, if they are working properly!

Router0

```
Osaka_CentR0(config)#interface fa0/0.10
Osaka_CentR0(config-subif)#standby 10 ip 192.168.10.254
Osaka_CentR0(config-subif)#standby 10 priority 90
Osaka_CentR0(config-subif)#standby 10 preempt
Osaka_CentR0(config-subif)#exit
Osaka_CentR0(config-subif)#standby 20 ip 192.168.10.254
% Address 192.168.10.254 in group 10
Osaka_CentR0(config-subif)#standby 20 ip 192.168.20.254
Osaka_CentR0(config-subif)#standby 20 priority 90
Osaka_CentR0(config-subif)#
%HSRP-6-STATECHANGE: FastEthernet0/0.20 Grp 20 state Speak -> Standby
%HSRP-6-STATECHANGE: FastEthernet0/0.20 Grp 20 state Standby -> Active
Osaka_CentR0(config-subif)#standby 20 preempt
Osaka_CentR0(config)#interface fa0/0.30
Osaka_CentR0(config-subif)#standby 30 ip 192.168.30.254
Osaka_CentR0(config-subif)#standby 30 priority 90
Osaka_CentR0(config-subif)#
%HSRP-6-STATECHANGE: FastEthernet0/0.30 Grp 30 state Speak -> Standby
%HSRP-6-STATECHANGE: FastEthernet0/0.30 Grp 30 state Standby -> Active
Osaka_CentR0(config-subif)#standby 30 preempt
Osaka_CentR0(config-subif)#exit
Osaka_CentR0(config)#interface fa0/0.40
Osaka_CentR0(config-subif)#standby 40 ip 192.168.40.254
Osaka_CentR0(config-subif)#standby 40 priority 110
Osaka_CentR0(config-subif)#standby 40 preempt
Osaka_CentR0(config-subif)#exit
%HSRP-6-STATECHANGE: FastEthernet0/0.40 Grp 40 state Speak -> Standby
%HSRP-6-STATECHANGE: FastEthernet0/0.40 Grp 40 state Standby -> Active
Osaka_CentR0(config)#interface fa0/0.90
Osaka_CentR0(config-subif)#standby 90 ip 192.168.90.254
Osaka_CentR0(config-subif)#standby 90 priority 110
Osaka_CentR0(config-subif)#
%HSRP-6-STATECHANGE: FastEthernet0/0.90 Grp 90 state Speak -> Standby
%HSRP-6-STATECHANGE: FastEthernet0/0.90 Grp 90 state Standby -> Active
```

```
Osaka_CentR0(config-subif)#standby 90 preempt
Osaka_CentR0(config-subif)#exi
show standby brief
```

Router1

Enter configuration commands, one per line. End with CNTL/Z.

```
Osaka_CentR1(config)#interface fa0/0.10
Osaka_CentR1(config-subif)# standby 10 ip 192.168.10.254
Osaka_CentR1(config-subif)# standby 10 priority 110
Osaka_CentR1(config-subif)# standby 10 preempt
%HSRP-6-STATECHANGE: FastEthernet0/0.10 Grp 10 state Speak -> Standby
Osaka_CentR1(config-subif)#
%HSRP-6-STATECHANGE: FastEthernet0/0.10 Grp 10 state Standby -> Active
Osaka_CentR1(config-subif)#exi
Osaka_CentR1(config)#interface fa0/0.20
Osaka_CentR1(config-subif)# standby 20 ip 192.168.20.254
Osaka_CentR1(config-subif)# standby 20 priority 110
Osaka_CentR1(config-subif)#
%HSRP-6-STATECHANGE: FastEthernet0/0.20 Grp 20 state Speak -> Standby
Osaka_CentR1(config-subif)#standby 20 preempt
Osaka_CentR1(config-subif)#
%HSRP-6-STATECHANGE: FastEthernet0/0.20 Grp 20 state Standby -> Active
exi
Osaka_CentR1(config)#interface fa0/0.30
Osaka_CentR1(config-subif)# standby 30 ip 192.168.30.254
Osaka_CentR1(config-subif)# standby 30 priority 110
Osaka_CentR1(config-subif)# standby 30 preempt
Osaka_CentR1(config-subif)#
%HSRP-6-STATECHANGE: FastEthernet0/0.30 Grp 30 state Standby -> Active
Osaka_CentR1(config-subif)#exi
Osaka_CentR1(config)#interface fa0/0.40
Osaka_CentR1(config-subif)# standby 40 ip 192.168.40.254
Osaka_CentR1(config-subif)# standby 40 priority 90
Osaka_CentR1(config-subif)#standby 40 preempt
Osaka_CentR1(config-subif)#
Osaka_CentR1(config-subif)#exi
%HSRP-6-STATECHANGE: FastEthernet0/0.40 Grp 40 state Speak -> Standby
Osaka_CentR1(config)#interface fa0/0.90
Osaka_CentR1(config-subif)# standby 90 ip 192.168.90.254
Osaka_CentR1(config-subif)# standby 90 priority 90
Osaka_CentR1(config-subif)#standby 90 preempt
Osaka_CentR1(config-subif)#exi
```

15

write memory

16 test

1.Accessing to SVI for management by ssh and login as a administrator.

```
C:\>ssh -l Admin1 192.168.90.248
```

Password:

```
Osaka_CentASW1#exi
```

2.Ping from PC in Vlan10 to PC in Vlan 40

```
C:\>ping 192.168.40.10
```

Pinging 192.168.40.10 with 32 bytes of data:

Request timed out.

Request timed out.

Reply from 192.168.40.10: bytes=32 time<1ms TTL=127

Reply from 192.168.40.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.40.10:

Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),

Approximate round trip times in milli-seconds:

Minimum = 0ms, Maximum = 0ms, Average = 0ms

3.Shutdown the interface of R2 and check if all vlan states become active at R1. After that no shutdown the interface of R2 and check ifl vlan states become standby and active.

R2

```
Osaka_CentR1(config)#interface fa0/0
```

```
Osaka_CentR1(config-if)#shutdown
```

R1

```
Osaka_CentR0#show standby brief
```

P indicates configured to preempt.

|

Interface	Grp	Pri	P	State	Active	Standby	Virtual IP
-----------	-----	-----	---	-------	--------	---------	------------

Fa 10	90	P	Active	local	unknown	192.168.10.254
-------	----	---	--------	-------	---------	----------------

Fa 20	90	P	Active	local	unknown	192.168.20.254
-------	----	---	--------	-------	---------	----------------

Fa 30	90	P	Active	local	unknown	192.168.30.254
-------	----	---	--------	-------	---------	----------------

Fa 40	110	P	Active	local	192.168.40.253	192.168.40.254
-------	-----	---	--------	-------	----------------	----------------

Fa 90	110	P	Active	local	192.168.90.253	192.168.90.254
-------	-----	---	--------	-------	----------------	----------------

```
Osaka_CentR0#
```

R2

```
Osaka_CentR1(config)#interface fa0/0
```

```
Osaka_CentR1(config-if)#no shut
```

R1

```
Osaka_CentR0#show standby bri
```

P indicates configured to preempt.

|

Interface	Grp	Pri	P	State	Active	Standby	Virtual IP
-----------	-----	-----	---	-------	--------	---------	------------

Fa 10	90	P	Standby	192.168.10.253	local	192.168.10.254
-------	----	---	---------	----------------	-------	----------------

Fa 20	90	P	Standby	192.168.20.253	local	192.168.20.254
-------	----	---	---------	----------------	-------	----------------

```
Fa 30 90 P Standby 192.168.30.253 local 192.168.30.254
Fa 40 110 P Active local 192.168.40.253 192.168.40.254
Fa 90 110 P Active local 192.168.90.253 192.168.90.254
Osaka_CentR0#
```

4.shut down the trunk ports of the root switch3 and check if the switch2 became root bridge.
Then no shutdown the trunk ports of switch 3 and check if the switch3 became root bridge

switch 3

```
Osaka_CentASW3(config)#interface range GigabitEthernet0/1 - 2
Osaka_CentASW3(config-if-range)#shutdown
Osaka_CentASW3(config)#interface fa0/1
Osaka_CentASW3(config-if)#shutdown
```

switch2

```
Osaka_CentASW2#show spanning-tree vlan 40,90
VLAN0040
Spanning tree enabled protocol ieee
Root ID Priority 28712
Address 0001.431B.C893
This bridge is the root
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Bridge ID Priority 28712 (priority 28672 sys-id-ext 40)
Address 0001.431B.C893
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time 20
Interface Role Sts Cost Prio.Nbr Type
-----
```

```
Fa0/1 Desg FWD 19 128.1 P2p
Gi0/2 Desg FWD 4 128.26 P2p
VLAN0090
Spanning tree enabled protocol ieee
Root ID Priority 28762
Address 0001.431B.C893
This bridge is the root
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Bridge ID Priority 28762 (priority 28672 sys-id-ext 90)
Address 0001.431B.C893
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time 20
Interface Role Sts Cost Prio.Nbr Type
-----
```

```
Fa0/1 Desg FWD 19 128.1 P2p
Gi0/2 Desg FWD 4 128.26 P2p
```

Switch3

```
Osaka_CentASW3(config)#interface range GigabitEthernet0/1 - 2
```

```
Osaka_CentASW3(config-if-range)#no shutdown
Osaka_CentASW3(config)#interface fa0/1
Osaka_CentASW3(config-if)#no shu
```

```
switch2
VLAN0040
Spanning tree enabled protocol ieee
Root ID Priority 24616
Address 0040.0B02.4675
Cost 4
Port 25(GigabitEthernet0/1)
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Bridge ID Priority 28712 (priority 28672 sys-id-ext 40)
Address 0001.431B.C893
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time 20
Interface Role Sts Cost Prio.Nbr Type
```

```
-----
Fa0/1 Desg FWD 19 128.1 P2p
```

```
Gi0/1 Root LSN 4 128.25 P2p
```

```
Gi0/2 Desg LSN 4 128.26 P2p
```

```
VLAN0090
```

```
Spanning tree enabled protocol ieee
```

```
Root ID Priority 24666
```

```
Address 0040.0B02.4675
```

```
Cost 4
```

```
Port 25(GigabitEthernet0/1)
```

```
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
```

```
Bridge ID Priority 28762 (priority 28672 sys-id-ext 90)
```

```
Address 0001.431B.C893
```

```
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
```

```
Aging Time 20
```

```
Interface Role Sts Cost Prio.Nbr Type
```

```
-----
Fa0/1 Desg FWD 19 128.1 P2p
```

```
Gi0/1 Root LSN 4 128.25 P2p
```

```
Gi0/2 Desg FWD 4 128.26 P2p
```